



MINOR PROJECT - II

ON

Co Work

(MCA – II SEMESTER; BATCH 2025-27)



Project Mentors

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Problem Description

- There are no platforms that aggregate both the CoWorking Spaces with the people who want them in single platform with the open competition between different space providers for real price discovery.
- Our project aims to bridge this gap in the market and to provide consumers with a easy to use and convenient way to book and facilitate a space to work. Also, our application provides an easy and convenient way to manage their day to day operations.

Aim and Objectives

- **Aim:**

- To develop a robust web application that aggregate CoWorking Spaces Market using ReactJS, Java's SpringBoot, MongoDB.

- **Objectives:**

- To allow users to search, select and discover Co Working Spaces around them
- To see the review/ratings and to review/rate Spaces
- Enable users to add smaller services in the form of Add-ons before or during their allotted time.

Methodology and Technology Used

- **Methodology used for Project Development:**

- **Stakeholder Mapping:** Identify needs for both "Space Seekers" (consumers) and "Space Providers" (businesses).
- **Feature Prioritization:** Define core functionalities like real-time search, competitive bidding/pricing, and booking management.
- **Data Modeling:** Determine the document structure for MongoDB to handle varied amenities and dynamic pricing.

Methodology and Technology Used

- **Methodology used for Project Development:**

- **Microservices Architecture:** Decouple the user-facing marketplace from the provider-focused management ERP.
- **API Strategy:** Design RESTful endpoints in Spring Boot for seamless communication between the React frontend and MongoDB.
- **Security Framework:** Implement JWT-based authentication and Role-Based Access Control (RBAC) to separate consumer and provider data.

Methodology and Technology Used

- Technology used for Project Development:
 - Java (JDK 21) – Core Backend programming language
 - SpringBoot (4.0.6) – Backend
 - MongoDB Atlas – Cloud based DBMS
 - ReactJS – Frontend
 - IDEs used: IntelliJ / VSCodiums

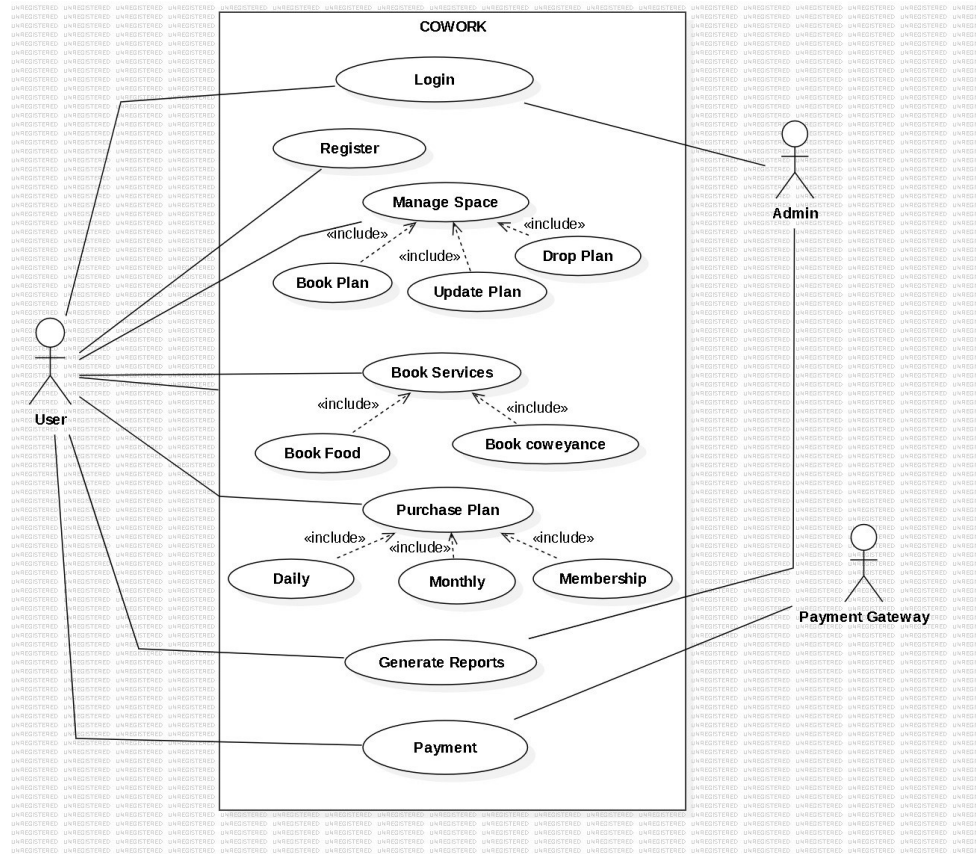
Project Modules

Project Modules:

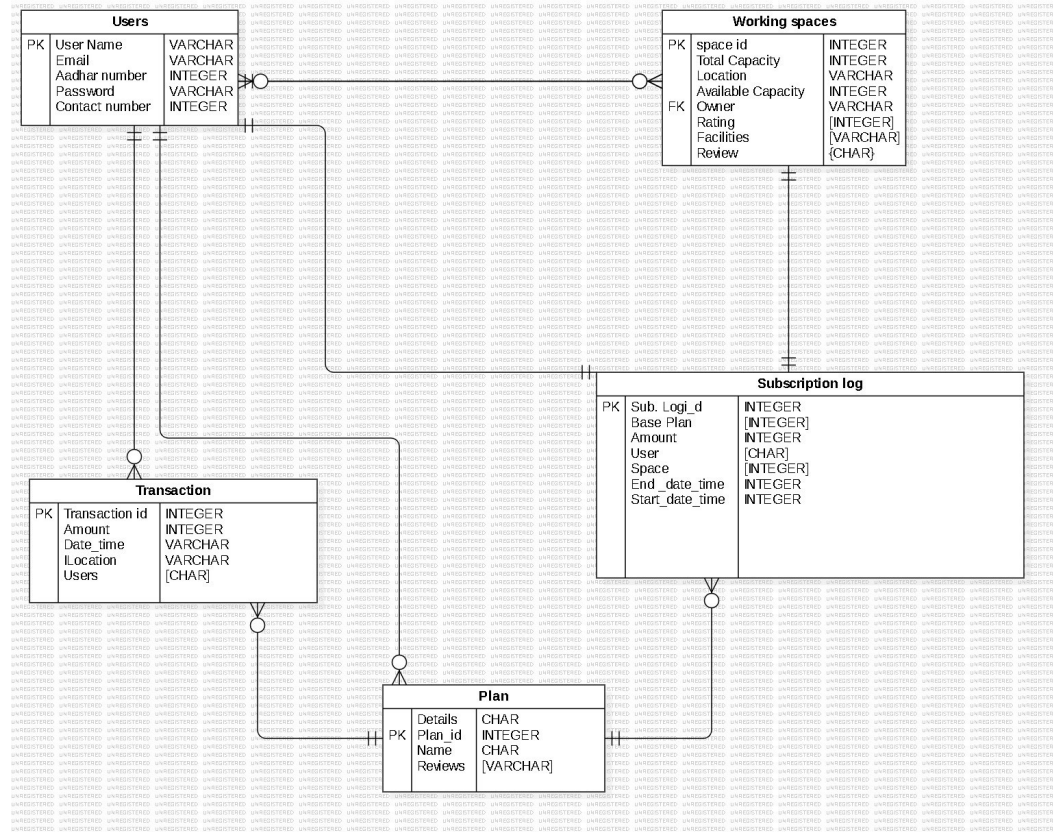
- **User Authentication & Profile Module:** Gateway to the application, managing identities and security.
- **Marketplace & Discovery Module:** "Search Engine" of the platform. (GeoSpatialSearch, Filtering Engine)
- **Booking & Transaction Module:** Handles Real-Time Availability, Pricing Engine and Payment Transactions.
- **Provider Operations (ERP) Module:** The internal tool for space owners to manage their business. (Inventory Management, Operational Dashboard, Maintenance Tracking)

System Design

Use Case Diagram



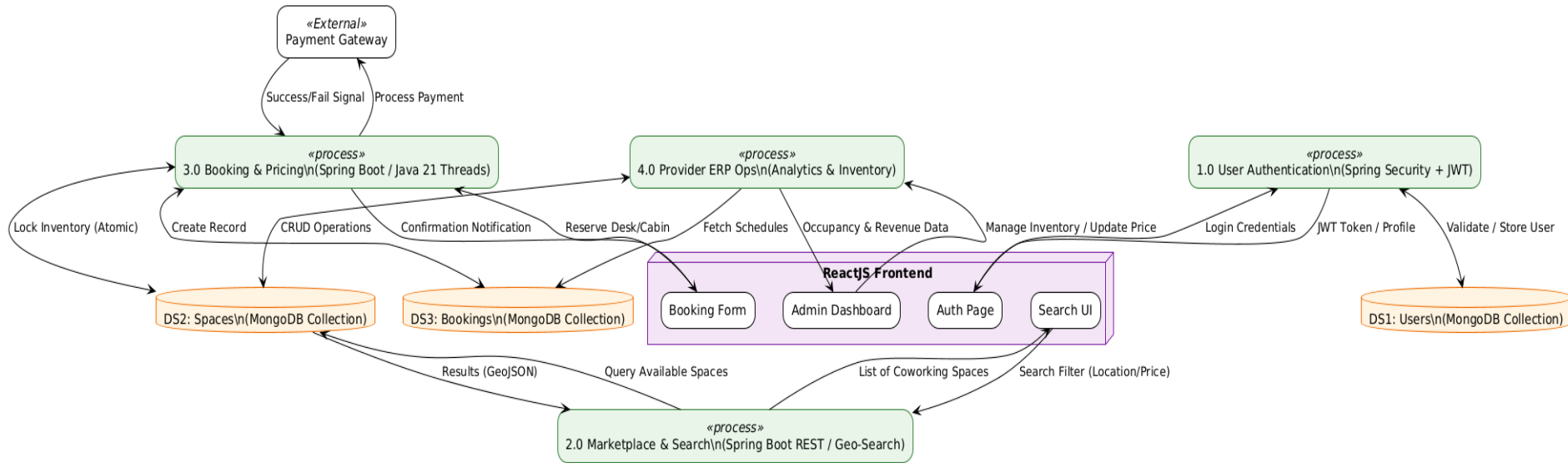
Entity-Relationship (ER) Diagram



Data Flow Diagram (Level - 0)



Data Flow Diagram (Level - 1)

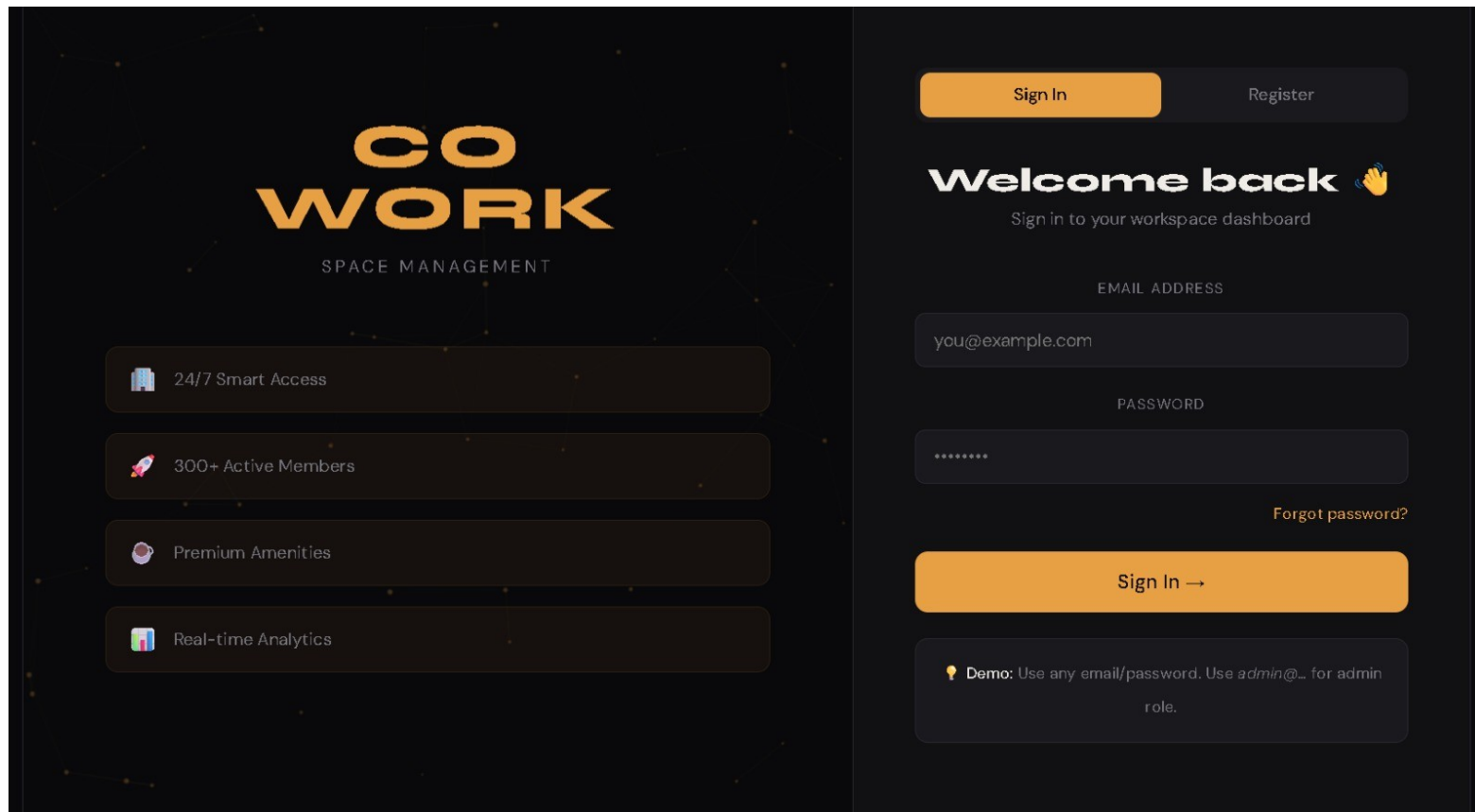


Operating Structure of Project

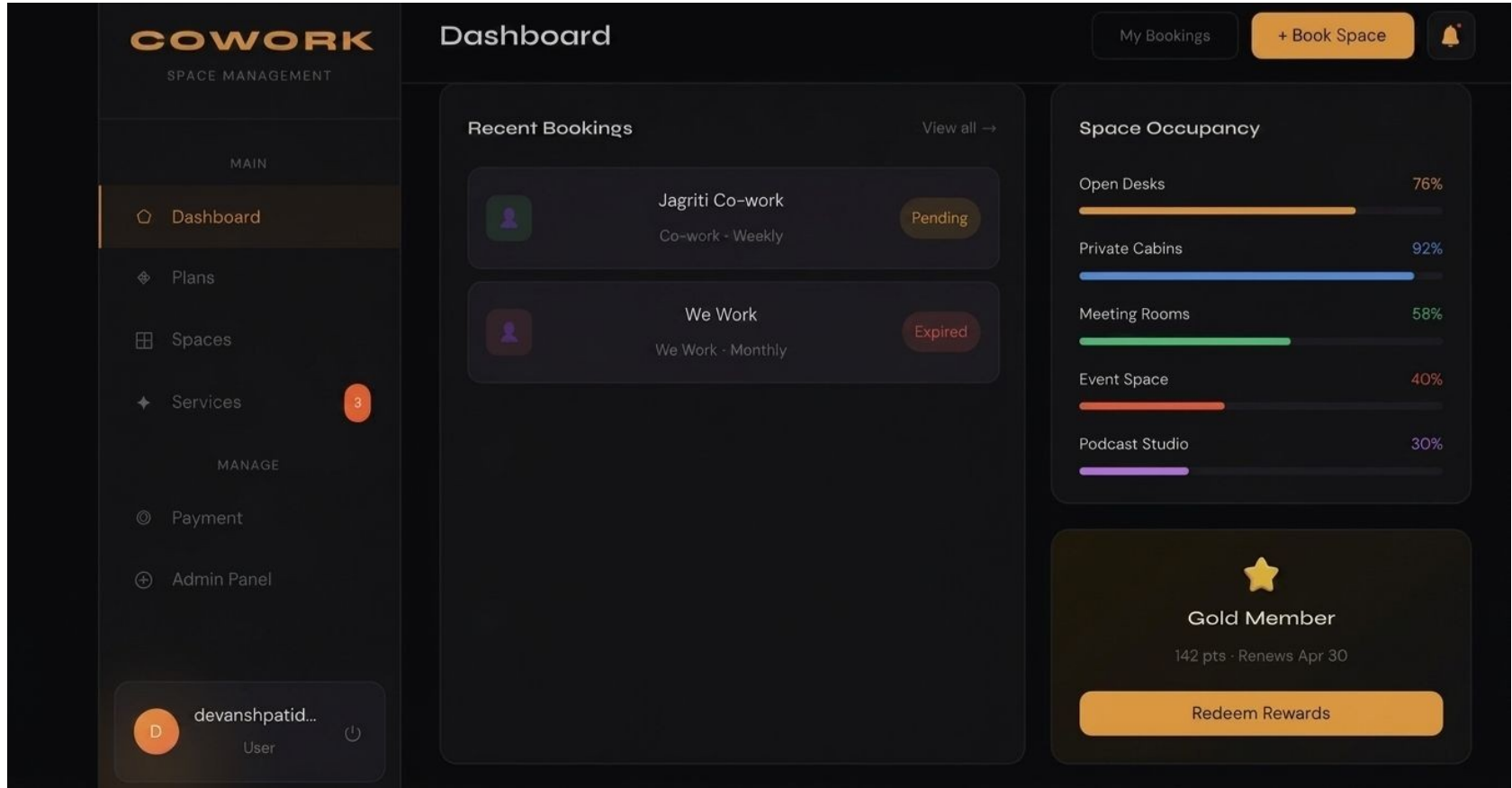
- **API Interaction:** React sends asynchronous HTTP requests via Axios/Fetch to Spring Boot REST endpoints.
- **Data Processing:** Spring Boot handles business logic and interacts with MongoDB, returning data in JSON format.
- **Dynamic Rendering:** React receives JSON responses to update the UI and manage state without refreshing.

Screenshots

Screenshot - 1



Screenshot - 2



The screenshot displays the COWORK SPACE MANAGEMENT Dashboard. The interface is divided into a sidebar, a main content area, and a top navigation bar.

Top Navigation Bar: Includes "My Bookings", "+ Book Space", and a notification bell icon.

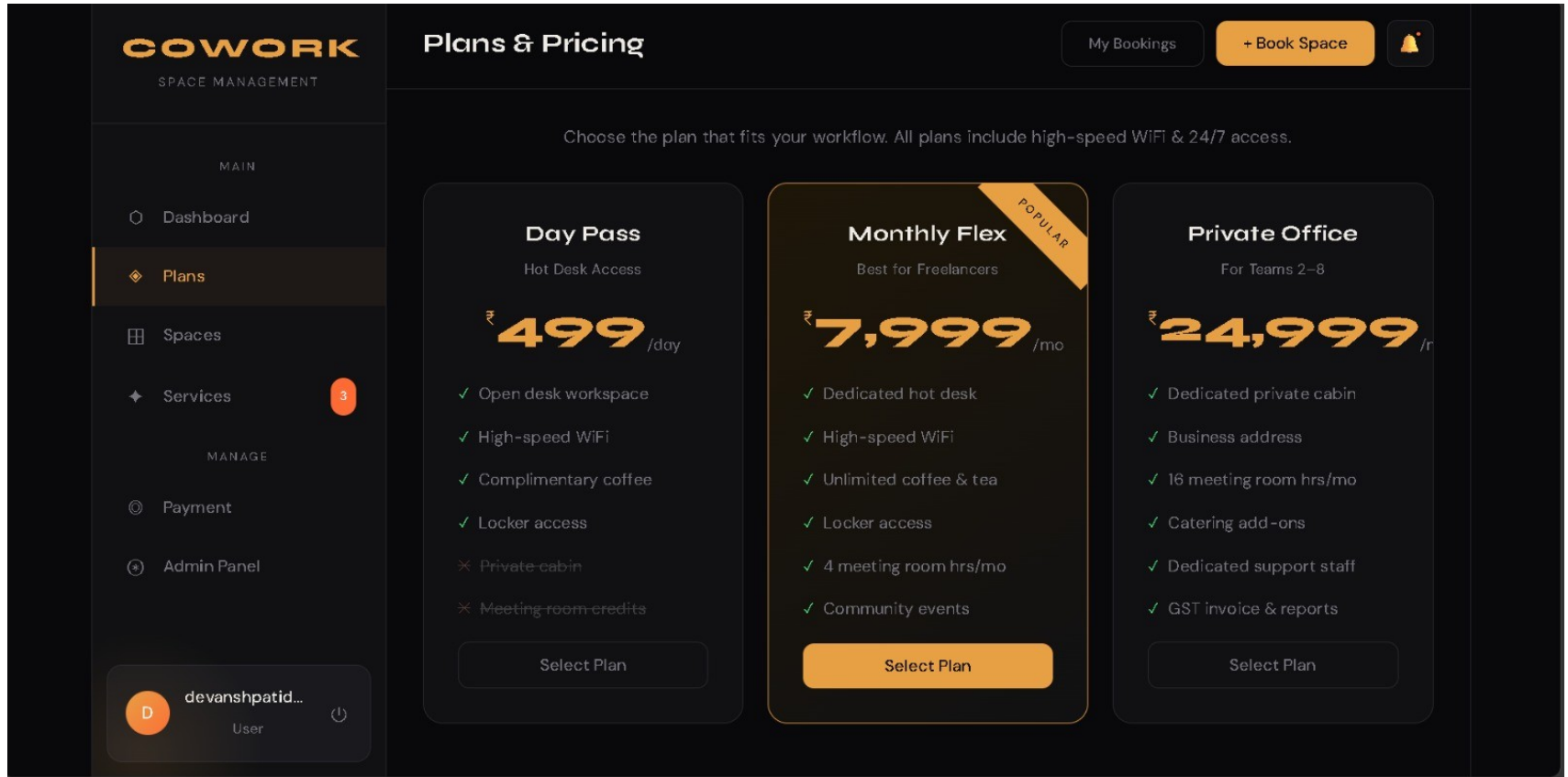
Sidebar: Contains the COWORK logo and "SPACE MANAGEMENT". Under the "MAIN" section, there are links for "Dashboard" (active), "Plans", "Spaces", "Services" (with a red badge showing "3"), and "MANAGE" (with sub-links for "Payment" and "Admin Panel"). At the bottom of the sidebar is a user profile for "devanshpatid..." with a "User" role and a power icon.

Main Content Area: Titled "Dashboard", it features two primary sections:

- Recent Bookings:** A list of bookings with a "View all →" link. It shows two entries:
 - Jagriti Co-work:** Co-work - Weekly, status "Pending" (yellow badge).
 - We Work:** We Work - Monthly, status "Expired" (red badge).
- Space Occupancy:** A section showing occupancy percentages for various spaces, each with a corresponding progress bar:
 - Open Desks: 76%
 - Private Cabins: 92%
 - Meeting Rooms: 58%
 - Event Space: 40%
 - Podcast Studio: 30%

Gold Member Section: Located at the bottom right, it features a star icon, the text "Gold Member", "142 pts - Renews Apr 30", and a "Redeem Rewards" button.

Screenshot - 3



COWORK
SPACE MANAGEMENT

Plans & Pricing

Choose the plan that fits your workflow. All plans include high-speed WiFi & 24/7 access.

Day Pass
Hot Desk Access
₹ **499** /day

- ✓ Open desk workspace
- ✓ High-speed WiFi
- ✓ Complimentary coffee
- ✓ Locker access
- ✗ Private cabin
- ✗ Meeting room credits

Select Plan

Monthly Flex
Best for Freelancers
₹ **7,999** /mo

- ✓ Dedicated hot desk
- ✓ High-speed WiFi
- ✓ Unlimited coffee & tea
- ✓ Locker access
- ✓ 4 meeting room hrs/mo
- ✓ Community events

Select Plan

Private Office
For Teams 2-8
₹ **24,999** /r

- ✓ Dedicated private cabin
- ✓ Business address
- ✓ 16 meeting room hrs/mo
- ✓ Catering add-ons
- ✓ Dedicated support staff
- ✓ GST invoice & reports

Select Plan

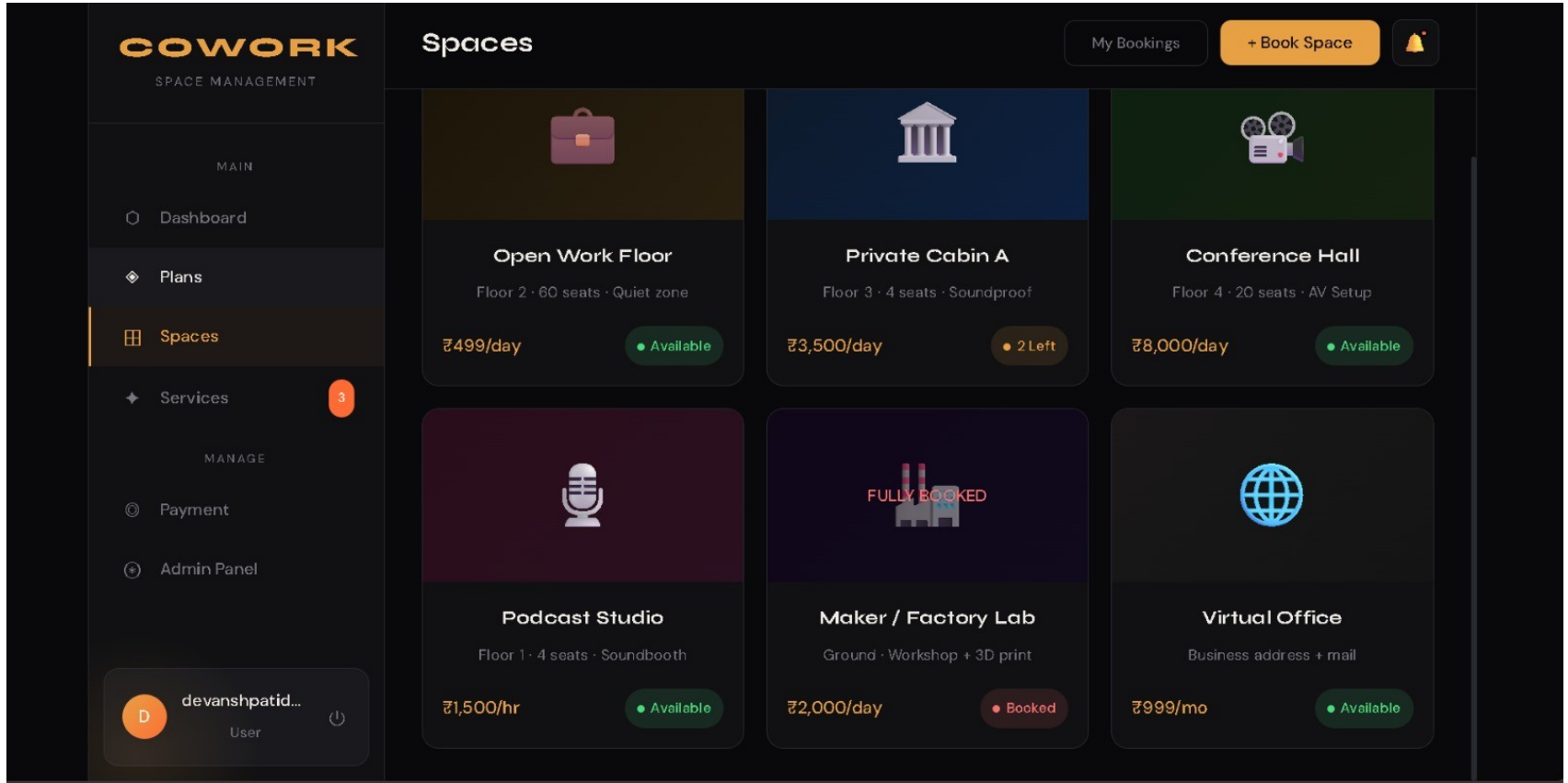
My Bookings + Book Space

Dashboard Plans Spaces Services 3

Payment Admin Panel

devanshpatid... User

Screenshot - 4



The screenshot displays the COWORK Space Management interface. On the left is a sidebar with navigation options: MAIN (Dashboard, Plans, Spaces, Services), and MANAGE (Payment, Admin Panel). The 'Spaces' section is active. The main area shows a grid of six space options:

Space Name	Location & Features	Price	Status
Open Work Floor	Floor 2 · 60 seats · Quiet zone	₹499/day	Available
Private Cabin A	Floor 3 · 4 seats · Soundproof	₹3,500/day	2 Left
Conference Hall	Floor 4 · 20 seats · AV Setup	₹8,000/day	Available
Podcast Studio	Floor 1 · 4 seats · Soundbooth	₹1,500/hr	Available
Maker / Factory Lab	Ground · Workshop + 3D print	₹2,000/day	Booked
Virtual Office	Business address + mail	₹999/mo	Available

At the top right of the main area are links for 'My Bookings', '+ Book Space', and a notification bell. The bottom left of the sidebar shows the user profile for 'devanshpatid...' with a 'User' role and a power button icon.

Screenshot - 5

COWORK
SPACE MANAGEMENT

MAIN

Dashboard

Plans

Spaces

Services 3

MANAGE

Payment

Admin Panel

devanshpatid...

User

Services & Add-ons

My Bookings

+ Book Space

**Food & Beverages**
Catering, snacks, meals
on demand
✓ Added

**Conveyance**
Cab booking, airport
transfers

**IT Support**
Tech setup, printer,
projector

**Secure Locker**
Daily or monthly locker
rental
✓ Added

**Event Hosting**
Host meetups,
workshops, launches

**Print & Scan**
Color / B&W printing,
scanning

**Mail Handling**
Receive and forward
courier

**Membership Perks**
Discounts, partner
benefits

Active Add-ons

SERVICE	PLAN	COST	STATUS
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Screenshot - 6

COWORK
SPACE MANAGEMENT

MAIN

○ Dashboard

◆ Plans

☐ Spaces

◆ Services 3

MANAGE

Ⓢ Payment

Ⓜ Admin Panel

D devanshpatid...
User

⏻

Make Payment

My Bookings + Book Space 🔔

Payment Method

Card

UPI

Net Banking

Wallet

CARDHOLDER NAME

Rahul Sharma

CARD NUMBER

4242 4242 4242 4242

EXPIRY CVV

MM/YY ...

Pay ₹8,499 →

Order Summary

Monthly Flex Plan ₹7,999

Food Add-on ₹250

Secure Locker ₹500

Member Discount - ₹250

Total ₹8,499

★ Gold Member

You'll earn 84 pts for this purchase

Future Scope

- AI Pricing Optimization: Predictive analytics for real-time rate discovery.
- Mobile Expansion: Native React Native apps for on-the-go booking.
- IoT Smart Access: Automated door locks and energy management.
- AR Workspace Tours: Immersive 3D virtual space previews.